PETRA ZLATARIĆ-NOVOSEL

Machine Learning Engineer

Osijek, Croatia | petra.zlataric@example.com | +385 55 555 0606 | github.com/pzn-ml-example

PROFESSIONAL SUMMARY

Machine learning engineer with more than five years of experience in natural language processing and document classification, from research and prototyping to production models. Experienced in transformer fine-tuning, model serving and production monitoring on Azure. Doctoral researcher in Croatian language processing.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | Slavonica Document AI d.o.o. | Osijek
03/2023–Present

- Developed an automated document classification model for public administration clients, achieving 94% accuracy across 23 classes, including invoices, contracts and official decisions.
- Fine-tuned BERTić and XLM-R transformer models for named entity recognition in Croatian documents.
- Introduced MLflow for experiment tracking and the model registry; models are delivered as FastAPI services in Docker containers on Azure ML and App Service.
- Established production model quality monitoring, including data drift monitoring and weekly reporting.
- Mentor two student interns.

Data Scientist | Baranjska kreditna banka d.d. | Zagreb, remote from Osijek
09/2021–02/2023

- Built customer churn prediction models using gradient boosting and LightGBM for the sales team.
- Classified topics in customer complaint texts using TF-IDF and logistic regression, later using fastText.
- Produced SQL reports using an Oracle data warehouse.

Research Project Assistant | Faculty of Applied Informatics, University of Drava
10/2020–08/2021

- Built and annotated a corpus of Croatian legal texts and wrote Python scripts.

EDUCATION

Doctoral Studies in Computer Science | University of Drava | 2022–Present
Research topic: Named entity recognition in Croatian legal texts

Master of Computer Engineering (Mag. ing. comp.) | University of Drava | 2020

SKILLS

Python, PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, fastText, spaCy, pandas, NumPy, SQL, MLflow, FastAPI, Docker, Azure ML, Git, Linux

LANGUAGES

Croatian: Native | English: C1 | German: B2

PUBLICATIONS

Zlatarić-Novosel, P. et al. (2024). Named entity recognition for Croatian administrative documents. Workshop on Slavic language processing (fictional proceedings).